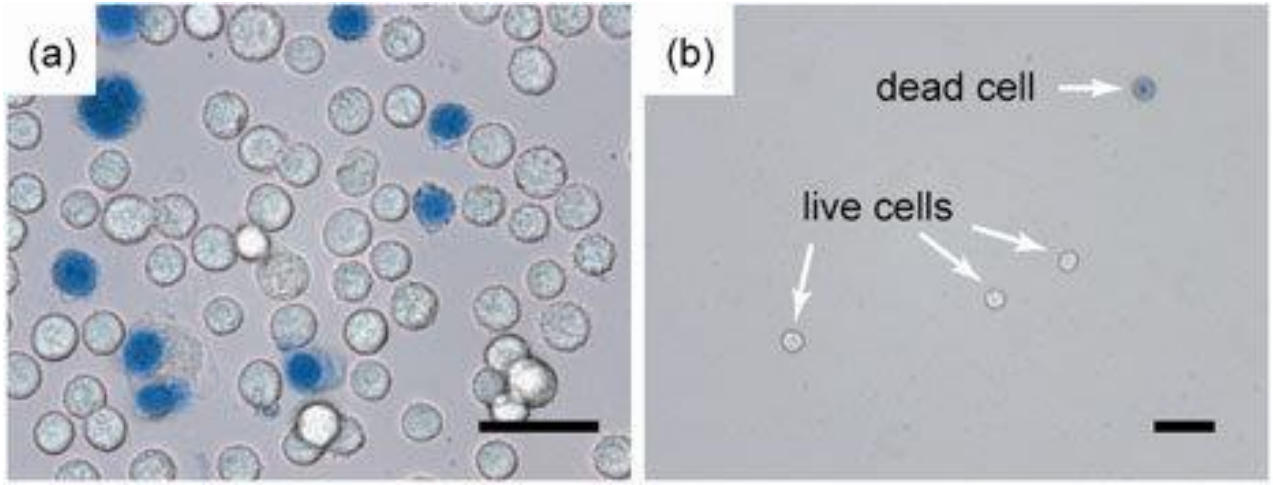
Welcome to Biol340
Practical Labs



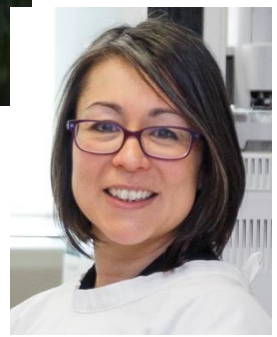
BIOL340 Staff

Coordinator:

- Dr Nikolas (Nik) Johnston

Lecturers:

- Dr Jody Gorman
- Prof Lezanne Ooi
- Dr. Dzung Do-Ha



Technical Staff:

- Dr. Mitchell St Claire-Glover

General enquiries email:
biol340-support@uow.edu.au

Specific lecturer emails can be found on the moodle site

Subject discussion forums available on moodle

What to do in an emergency?

KEEP CALM – STAY SAFE

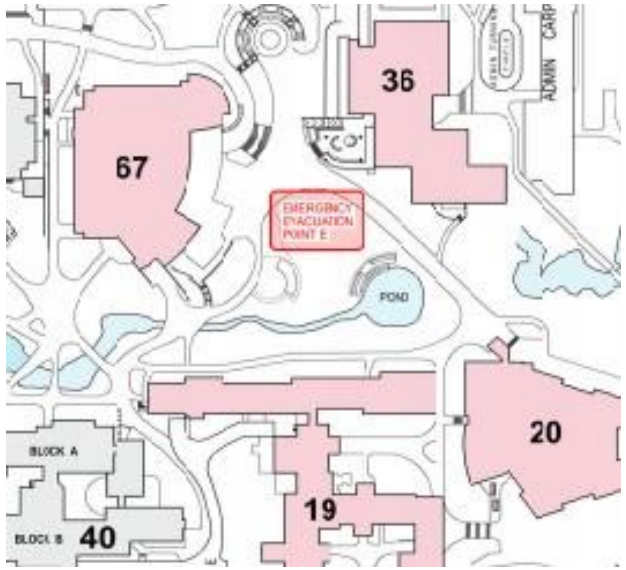
If the alarm sounds or you are notified to evacuate:

- Follow instructions of building warden or staff member
- Leave by the nearest safe emergency exit
- Proceed to your emergency evacuation assembly point
- Await further instructions
- Do not return to the building until it is safe to do so

If required to take shelter:

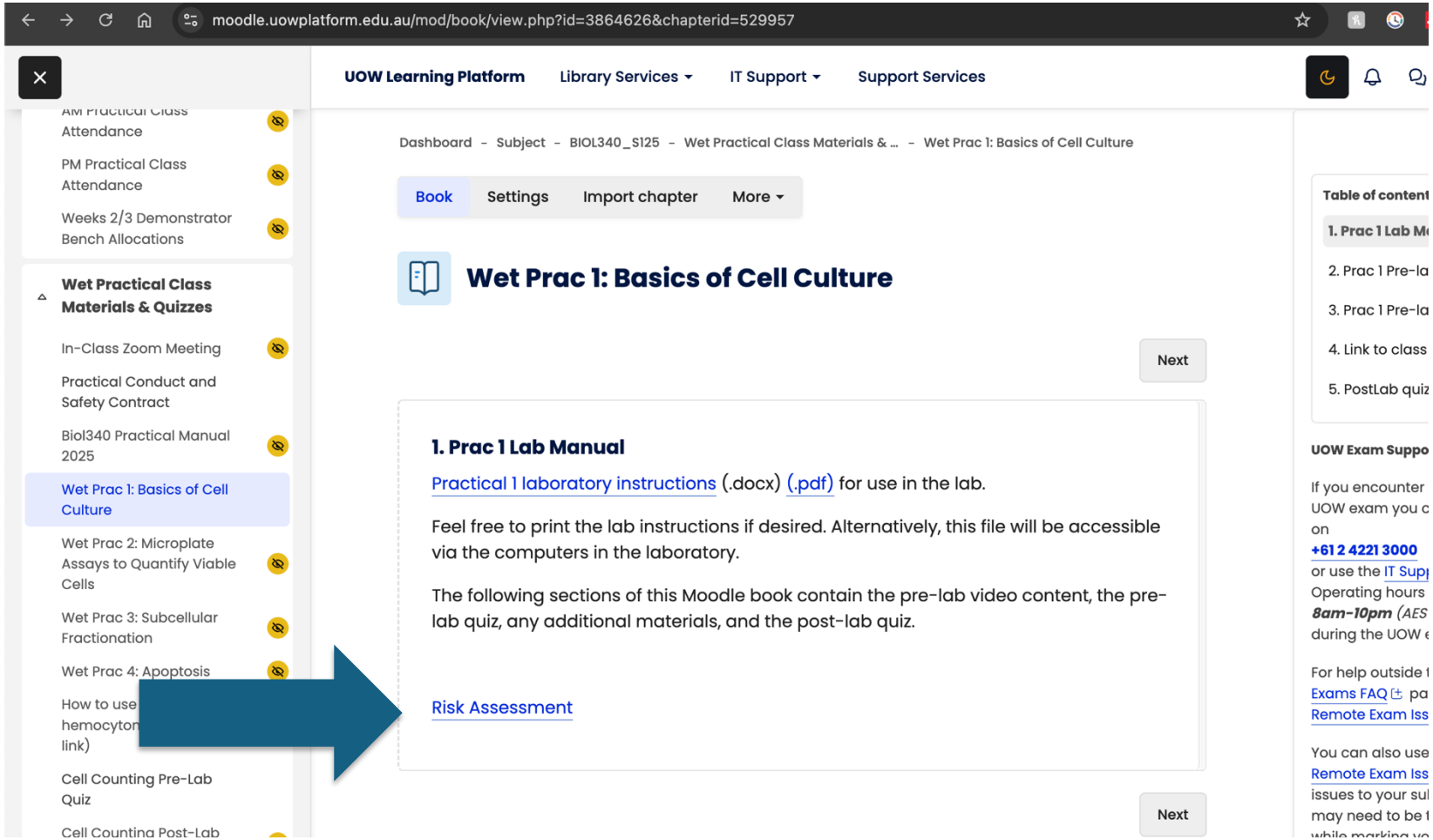
- Follow instructions of building warden or staff member
- Lock doors, close windows/blinds and seek refuge
- Await further instructions

The nearest assembly area for building 20 (lectures) is:



The nearest assembly area for building 43 (pracs) is:

Lab Safety


moodle.uowplatform.edu.au/mod/book/view.php?id=3864626&chapterid=529957
UOW Learning Platform
Library Services
IT Support
Support Services
AM Practical Class Attendance
PM Practical Class Attendance
Weeks 2/3 Demonstrator Bench Allocations
Wet Practical Class Materials & Quizzes
In-Class Zoom Meeting
Practical Conduct and Safety Contract
Biol340 Practical Manual 2025
Wet Prac 1: Basics of Cell Culture
Wet Prac 2: Microplate Assays to Quantify Viable Cells
Wet Prac 3: Subcellular Fractionation
Wet Prac 4: Apoptosis
Cell Counting Pre-Lab Quiz
Dashboard - Subject - BIOL340_S125 - Wet Practical Class Materials & ... - Wet Prac 1: Basics of Cell Culture
Book
Settings
Import chapter
More
Wet Prac 1: Basics of Cell Culture
Next
1. Prac 1 Lab Manual
Practical 1 laboratory instructions (.docx) (.pdf) for use in the lab.
Feel free to print the lab instructions if desired. Alternatively, this file will be accessible via the computers in the laboratory.
The following sections of this Moodle book contain the pre-lab video content, the pre-lab quiz, any additional materials, and the post-lab quiz.
Risk Assessment
Next
Table of content
1. Prac 1 Lab M
2. Prac 1 Pre-la
3. Prac 1 Pre-la
4. Link to class
5. PostLab quiz
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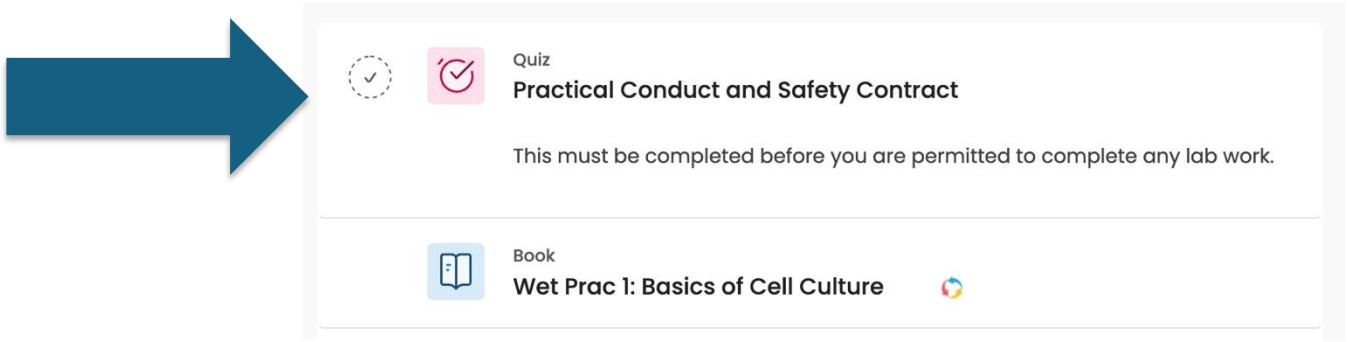
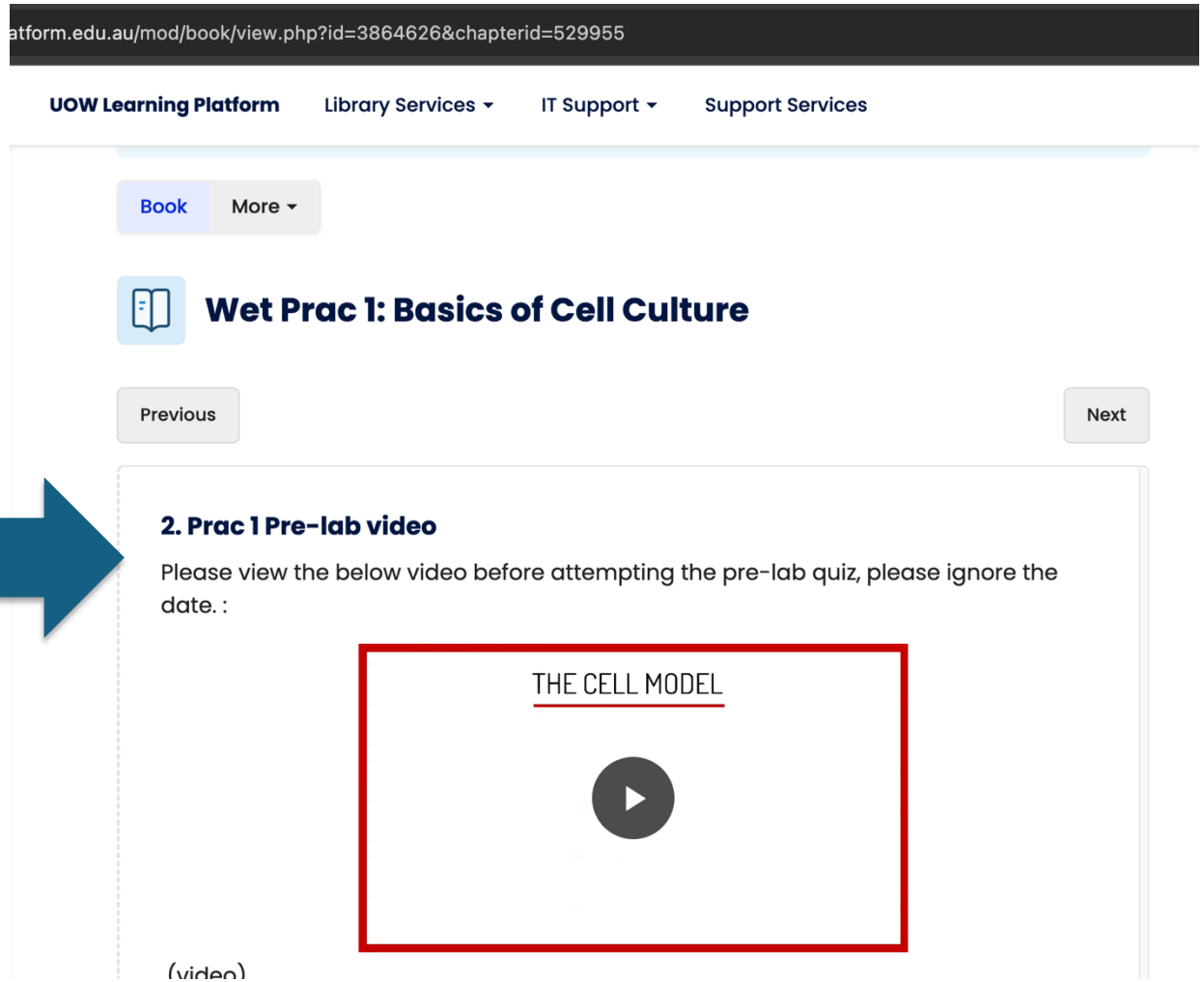
Lab Safety

Safety contract in Week 1 folder



Please complete now

Lab Safety



After week 1 must be complete prior to labs

Biol340 Practical Project

Testing the efficacy of a new drug against a mitochondrial disease

Over the next few weeks you will be working on a group project across both the wet and dry labs

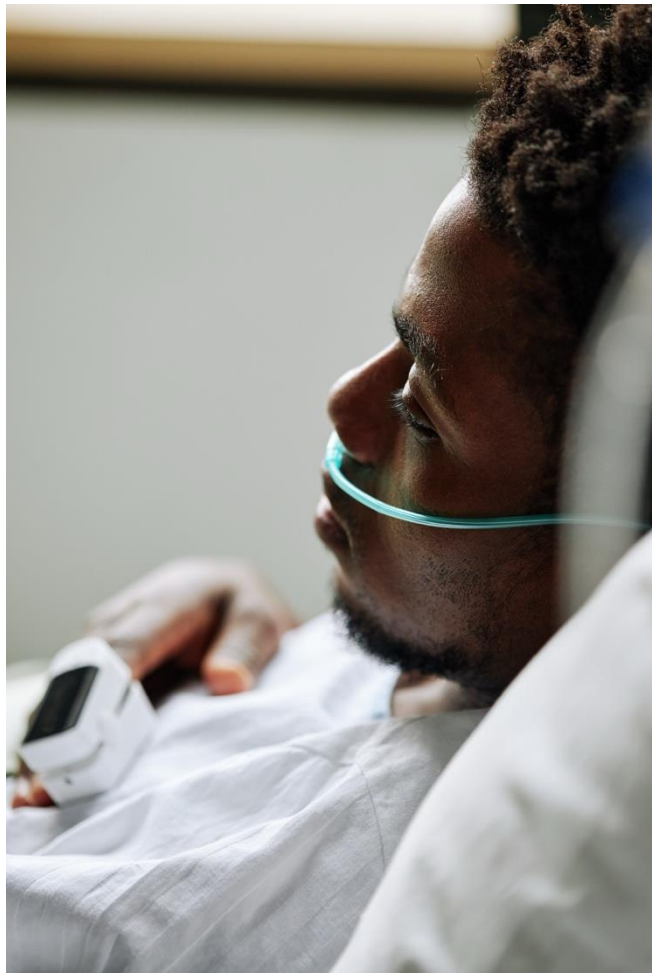
We will be determining the efficacy and action of a new drug to tream mitochondrial diseases

Mitochondrial diseases are a group of genetic disorders affect how cells are able to produce energy and can be life threatening.

They can include symptoms such as:

- Lactic acidosis
- Seizures
- Sudden vision loss
- Diabetes

The disease is of genetically inherited and current treatments are limited which restricts options for affected individuals

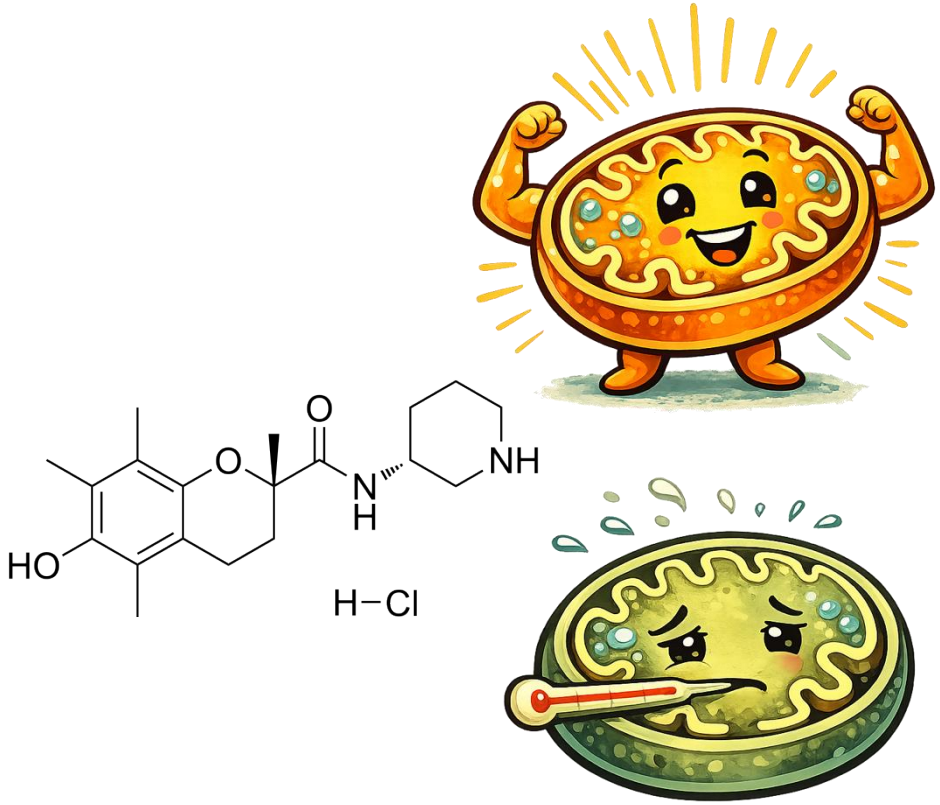


Biol340 Practical Project

Testing the efficacy of a new drug against a mitochondrial disease

Project Background:

- There are very few drug treatments for mitochondrial diseases
- A pharmaceutical company has developed a putative drug called Respiranox^{tm} which is a variant of sonlicromanol[1]
- This drug has had success in improving mitochondrial function in-vitro mouse cell experiments.
- Your job will be to cell biology techniques to test the efficacy and action of Respiranox^{tm} on a diseased human cell line.



[1]Smeitink et al. Brain, 2025 https://doi.org/10.1093/brain/awae277

Biol340 Practical Project timeline

To determine the efficacy and action of Respiranox^{tm} we will use the following plan:

Wet labs

WEEK 1:

Use **Cell counting** techniques to characterise a human cell line affected with a mitochondrial disease

Week 2:
Use **Microplate assays to t**est the efficacy of the drug treatment on this cell line and determine optimal concentration.

Week 4:

Use **subcellular fractionation** techniques to determine the action of the drug is specific to mitochondria

Dry labs

Week 3:

Analysis of class drug treatment data set using big-data analysis methods. Including R and data visualisation

+

Week 5:
Continued analysis and report writing on drug efficacy results

Techniques You'll Use

Microscopy – Visualizing cellular structures – Wet Lab 1

Microplate assays – Measuring drug efficacy– Wet Lab 2

Spectrophotometry – Assessing mitochondrial function – Wet Lab 3

Fluorescence microscopy & flow cytometry – Detecting apoptosis – Wet Lab 4

From Data to Scientific Report

After 3 wet lab experiments we will analyze results in R assisted by AI tools.

Interpret findings and compile them into a formal scientific report.

We will then refine this piece of work using generative AI and peer-feedback

You will be completing your project **in pairs** – including data analysis and report writing

The final assessment is individual

Practical Assessments

Practical assessments include:
Assessment 3) four post-lab quizzes
Assessment 4) a group-based scientific report
Assessment 5) a final reflection and revision task completed individually.

Assessments are linked and are structured as a research project where you will complete a experiment, write up your results, receive feedback and then revise your writing. This mirrors the peer-review publication process.

A3: Post-lab quizzes

- Completed Individually
- These quizzes help frame the results and discussion that you will write up

A4: Scientific report

- Group-based
- This report formalises the work you completed in your practicals and is peer-assessed.

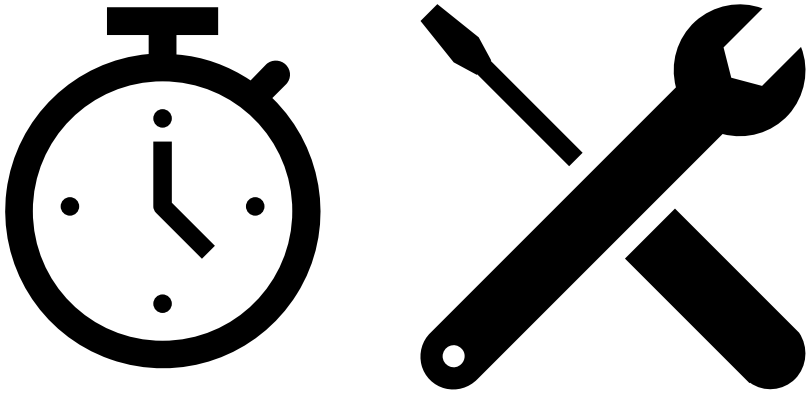
A5:GenAI revision & reflection

- Completed Individually
- Use feedback from peer-assessed report and Gen-AI tools to revise your writing.

Reasonable Adjustments

It is your responsibility to contact Student Accessibility, and me:

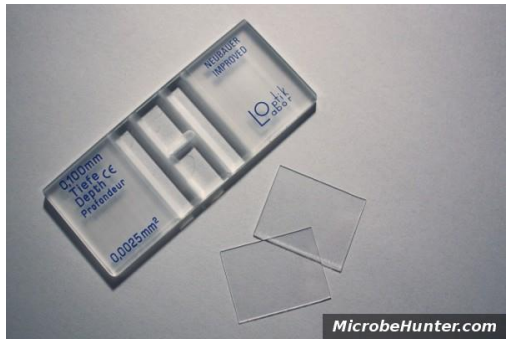
- The sooner the better
- Adjustments can be discussed
- Appropriate measures put in place asap

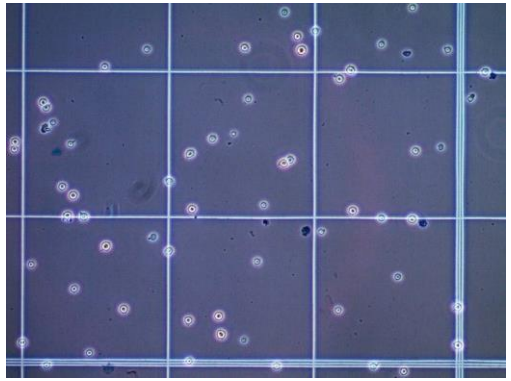


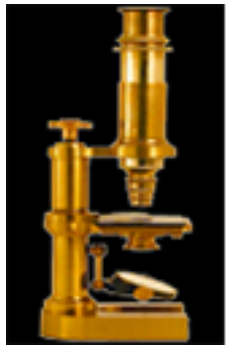
Today in Wet Lab 1

Objectives

1. Gain a basic appreciation of the techniques required to maintain cells in culture.
2. Learn the technique of cell counting using a haemocytometer.
3. Use image analysis techniques to complete automated cell counting.







ImageJ

Image Processing & Analysis in Java